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INVITED COMMENTARY

The Benefits of Research-Embedded Training Camps in Sport Sciences

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Background: Sport scientists aim to enhance athletic performance by developing, implementing, and evaluating evidence-based strategies. However, a common challenge to maximize sport science's impact is the limited involvement of "research end users" (ie, athletes, coaches, and practitioners) in contextualizing these strategies. To address this, research-embedded training camps (RETCs) offer a collaborative opportunity bringing together all stakeholders to tackle real-world challenges by blending scientific insight with practical application for better knowledge translation or production. ***Purpose:*** This invited commentary explores the behind-the-scenes aspects of conducting successful RETCs. We discuss opportunities and challenges involved, drawing on our practical experience with environmental stress interventions. ***Evidence:*** Implementing an RETC requires a tailored intervention, shaped by the athlete group, research questions, and available resources—essentially answering the *who*, *what*, *why*, *when*, *and where* questions. Key principles for success include an athlete-centered focus, a diverse multidisciplinary team, scientific rigor, balancing control with flexibility, and adaptability. A coconstructed design ensures that the intervention remains scientifically relevant and practical and directly addresses real-world needs of athletes. ***Conclusion:*** Through the lens of environmental stress interventions, RETCs should prioritize practicality, collaboration, and athlete-focused approaches. Conducting interventions *with* athletes rather than *on* them can significantly enhance performance and support evidence-based training strategies (knowledge translation) and/or provide new actionable insights (knowledge production).

Keywords: implementation research, athlete-centered approach, codesign, environmental stress, real-work challenges, research impact

As sport scientists, our intention is to provide expert advice and support to athletes, coaches, and practitioners—the "research end users"—to give them a competitive edge.[1] We strive to develop, implement, and evaluate evidence-based, quality-assured strategies that are both safe and effective in enhancing performance. But do we truly succeed? A recurring challenge is the limited involvement of these research end users in shaping interventions beyond the academic sphere,[2] despite the fact that their continuous engagement is crucial for fostering trust and collaboration in a real-world context.[3,4] Additionally, most sport-sciences research involves tier 0 to 2 participants, occasionally tier 3, but rarely tier 4 to 5.[5] Increased engagement with high-performance sport professionals is essential for driving research, designing studies, analyzing data, and applying findings to practice.[1,6]

Training camps may involve athletes temporarily relocating, typically to foreign venues, for supervised sessions lasting 10 days or more.[7] During these camps, sport scientists gather physical performance and wellness data to manage training loads, aid recovery, and assess both immediate and longer-term effects of interventions (eg, nutritional support[8] or specific "shock" microcycles[9]). Engaging end users in a culture of shared knowledge allows sport scientists and organizations to offer strong support while accessing elite athletes in top-notch facilities and in ecologically valid environments. Research-embedded training camps (RETCs) foster collaboration among athletes, coaches, and sport scientists, working together to tackle real-world challenges. This synergy enriches the experience for all involved while providing a platform to test evidence-based strategies (ie, knowledge translation) and/or produce new scientific insights (ie, knowledge production)—a true win-win.

Hypoxia and heat stress are key topics in sport sciences, crucial for acclimatization/acclimation and performance optimization. Many elite athletes incorporate altitude (ie, live high-train high, live high-train low, and live low-train high)[10] and/or heat[11] training camps into their annual programs. These interventions, whether applied alone or combined, have also been explored for improving performance in sea-level temperate environments, though the results remain mixed.[12]

Our intention is not to critique the value of ecologically valid evidence versus traditional academic research (for a review, see Brocherie and Beard[1]) but to delve into the practical aspects of applied intervention-style studies, specifically those involving RETCs. We aim to clarify the concept of RETCs, their potential benefits, and the challenges of implementing such interventions. In this invited commentary, we will draw on our experiences and published examples of RETCs involving environmental stressors (ie, altitude and/or heat training). However, the principles discussed are likely applicable to RETCs not focused on environmental stress, as well.

Why Do We Need RETCs?

Evidence-Based Decision Making and Real-World Challenges

Training camps offer valuable insights into the effectiveness and potential risks of interventions like altitude (ie, acute mountain

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sickness and deteriorated sleep)[10] and/or heat training (ie, hyperthermia, electrolyte imbalances, and dehydration)[11] in real-world contexts. These insights help research end users refine and scale-up evidence-informed practices in areas such as acclimatization/acclimation, hydration, nutrition, training, sleep monitoring, and safety protocols. Additionally, RETCs address real-world challenges, such as preparing for competitions in unfamiliar environments. For example, while altitude camps are common, global warming has shifted the focus toward incorporating heat acclimatization during tapering.[13] RETCs play a key role in addressing these challenges and exploring innovative solutions.[14]

Innovative Training Insights

RETCs offer unique opportunities to test and explore innovative training paradigms in realistic settings. Drawing on our research in repeated-sprint training in hypoxia, we introduced a novel approach to maximize the hypoxic stimulus (ie, simulating 5000-m altitude compared with the typical 3000 m) while preserving training quality through normoxic interset rest.[15] This approach was tested with world-class female rugby sevens players during a 9-day preparation camp for a World Rugby Sevens Series tournament.

Personalized Training

RETCs enable the study of individual training responses, allowing for personalized altitude or heat exposure protocols by determining and adjusting the optimal "dose" for each athlete. For instance, changes in hematocrit after a week of heat acclimatization can predict the benefits of short-term acclimatization for professional Australian football players.[16] These camps can track the time course of adaptations and individual responses, distinguishing between *slow* and *fast* responders heat exposure.[17] Personalization helps refine best practices for sport foods and/or dietary supplements[8] and cooling strategies[11] for competitions in extremes conditions. Additionally, RETCs enable support staff to conduct research to address performance limitations by integrating scientific findings with daily training, optimizing strategies for extreme environments. For instance, a 14-day "in-season" shock microcycle including 6 sessions of repeated-sprint training in hypoxia revealed significant effects on specific fitness and tennis performance in a rookie professional player.[18]

Collaboration and Knowledge Exchange

RETCs benefit greatly from fostering collaboration and knowledge exchange, which helps integrate research into practical training approaches.[19] Workshops offer a platform to share insights and lessons from practical applications of altitude (ie, *altitude training and team sports* conference)[10] and heat training (ie, *training and competing in the heat* conference).[11] These discussions lead to actionable insights for daily practice. However, some practitioners may be reluctant to share findings due to perceived conflicts of interest among elite organizations. Enhancing cooperation among competitors through the *coopetition* conceptual framework could drive innovation in elite sport.[20]

How Should RETCs Be Constructed for Maximized Impact?

Executing an RETC is not a *one-size-fits-all* process; it varies depending on research objectives, available resources, and team size. Effective implementation requires clear research motivations and close collaboration with research end users to ensure the process meets their needs.[19] Interventions and protocols should address relevant issues and focus on practical aspects that give athletes a competitive edge. Essential considerations are further discussed below.

Athletes' Preferences Are the Focal Point

High-quality scientific support must prioritize athletes' needs and minimize disruptions to their regular training. Incorporating athlete feedback on previous experiences with heat or hypoxia is crucial. For instance, while some athletes may be comfortable spending extended hours (~12–14 h/d) in a normobaric chamber, others may find it isolating. Specialized facilities like climatic chambers, hypoxic tents, or altitude hotels can facilitate controlled studies and even double blinding.[10] Preferences often vary based on training goals, budget, and logistics. Some athletes may prefer the immersive experience of real altitude, enjoying scenic, remote locations and focused training, while others may prioritize the convenience and control of simulated altitude environments, allowing them to experience hypoxic conditions (ie, altitude hotels) without the time and cost of travel. Coaches and sport scientists should use this feedback to refine environmental stressor delivery, ensuring the camp remains effective and athlete-friendly.

Programming Considerations

RETCs should be integrated into athletes' annual programming, unlike traditional intervention studies where participants often adjust their training to fit experimental requirements. Choosing the appropriate duration for training camps is crucial to balance meaningful results with teams' schedules and competitive commitments. Traditional altitude camps typically last 3 to 4 weeks, which may not always be feasible due to programming constraints. Therefore, elite team sports are increasingly exploring shorter options, such as 14-day live high-train low and high altitude training.[21] Similar considerations apply to heat acclimation, where prolonged interventions (>7–10 d) may be challenging during the regular season.[22] Research outputs from RETCs can also be limited by small sample sizes, but a multicentric approach—combining performance data from different teams and/or locations—can help address this issue. Collecting data over multiple years, as seen in studies on elite male rugby players assessing heat and altitude effects over 3 in-season training camps,[23,24] is another viable solution.

Multidisciplinary Team

Assembling a diverse team of sport scientists, coaches, nutritionists, physiologists, and medical experts likely ensure a holistic understanding of training camp effects and tailor interventions effectively.[25] Clearly defined roles and expectations foster a positive culture and enhance success. Including both experienced and junior researchers encourages collaboration and balances the workload. Industry-based work-integrated learning, such as cofunded PhD scholarships and research internships, can facilitate this.[26] Engaging all stakeholders during the planning phase ensures alignment of research goals with the needs and preferences of athletes and coaches, maximizing their involvement and commitment.

Balance Between Control and Flexibility

While controlling for extraneous and confounding variables is often challenging in research settings,[27] training camps offer the

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advantage of standardizing key factors, such as training loads, nutrition (including food, fluids, and supplementation), medication, sleep, biological rhythms, and recovery especially when services are centralized, as seen in altitude and/or heat RETCs at Aspetar Hospital.[15,21,23,24] However, not all aspects require strict control (eg, standardizing menstrual cycle phases may limit female team-sport athletes' participation and compromise ecological validity, as they must align with the competition calendar). Striking the right balance between research and training activities is key to avoid overburdening athletes (and staff), which could impact compliance and engagement, while still ensuring maximal control. For instance, in our aforementioned study on repeated-sprint training in hypoxia,[15] players underwent 5 cold water immersion recovery sessions over 9 days. While this may have influenced some outcomes, strict control over this modality might have diminished the practical relevance, as athletes typically use cold water immersion postsessions. Detailed records of daily training loads, menstrual status, and environmental conditions are essential for addressing peer-review inquiries and enriching data analysis by providing context. While organization is critical, flexibility is equally important. Testing or training at terrestrial altitude may encounter unforeseen issues, such as adverse weather. Having a contingency plan, such as securing an indoor venue, ensures smooth operations despite unexpected challenges. In brief, embrace the chaos and remain adaptable!

Consider the Benefit:Cost Ratio When Selecting Monitoring/Testing Tools

Implementing valid and reliable measures in applied settings is challenging due to costs, time, and resource limitations.[28] When selecting tools for monitoring altitude[10] and heat training[11], it is crucial to balance benefits with budget constraints and available resources, avoiding pseudoscience (ie, untested smart devices or wearables on the market).[29] While measuring hemoglobin mass via carbon monoxide rebreathing is essential for confirming hematological responses, it may not be necessary when performance is the primary focus.[30] Tools for assessing hypoxia or heat stress should be chosen based on their validity, reliability, and available resources (ie, budget, facilities, and staff). For example, gastrointestinal capsules can be a practical alternative to rectal temperature measurements for heat acclimation, if budget permits. Monitoring athlete wellness and fatigue is vital for managing environmental stress effectively. Despite some performance gains from short-term heat acclimation, such as repeated-sprint training in the heat (five 60-min sessions over 7 d at 40 °C and 40% relative humidity),

Table 1 Tips for Maximizing the Impact of a Research-Embedded Training Camp

Item	Recommended actions
Customized research design	Tailor the research design to suit the specific athlete cohort, the research questions you aim to answer, and the available resources. Consider the unique characteristics and needs of the athletes to ensure the study is relevant and applicable to them.
Comprehensive documentation	Thoroughly document all aspects of the research process, including the study design, data collection/analysis methods, athlete characteristics, and any adjustments made during the camp (although they should be kept minimal). Clear documentation is essential for transparency, reproducibility, and addressing potential reviewer queries.
Practical monitoring, scientifically validated	Monitor athletes' daily training loads, including physical and psychological aspects, using submaximal tests, validated questionnaires, or technology. This helps identify injury or burnout risks. For female athletes, tracking menstrual status and phase is essential for optimizing performance, recovery, and injury prevention. Additionally, monitoring sleep, diet, medications, and timing ensures a holistic approach, enabling tailored interventions.
Environmental conditions	Take note of environmental factors that might influence athlete performance, such as temperature, humidity, altitude, carbon dioxide, or pollution levels. These variables should be factored into the research design and data analysis.
Balanced training and research	Strive to strike the right balance between the demands of the research and the needs of the athletes. If the research requirements become too burdensome or intrusive, it may lead to decreased compliance and interest from the participants.
Cultivating a positive culture	Foster a positive and collaborative atmosphere between the research team and the athletes. Building trust and rapport enhances participant engagement, cooperation, and adherence to study protocols.
Effective communication	Provide feedback and relevant information to the athletes during the camp. Share expertise and findings with them so they can benefit from the knowledge gained and understand the significance of their involvement.
Mix of researchers	Assemble a research team that comprises both experienced and junior researchers. This not only distributes the workload more effectively but also allows for the transfer of knowledge and skill development among team members.
Data management and analysis	Implement robust data management protocols to safeguard the accuracy and integrity of the data. Analyze the data rigorously and consider involving statistical experts to ensure sound interpretation of the results.
Long-term follow-up	Consider the possibility of conducting long-term follow-up studies to assess the lasting impact of the camp and the effectiveness of any interventions.
Dissemination of results	When not restricted by embargo or cultural issues, share relevant research findings through publications, conferences, and other appropriate channels. Disseminating results contributes to the broader sport sciences' community and can also benefit other stakeholders by enhancing training practices and performance strategies.
Continuous improvement	Reflect on the intervention's strengths and limitations after its completion. Use this feedback to improve future iterations and optimize the research design for subsequent studies.

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residual muscle fatigue may necessitate additional recovery time between sessions.[31] Integrating research with training optimizes resources,[1] allowing for efficient data collection and immediate feedback for continuous adjustments. However, the fast-paced nature of this environment may limit the depth of analysis typically needed for more comprehensive studies, which often require larger data sets and advanced statistical evaluations.[32]

Practical Implications

Applying a codesigned approach, as recommended by Slattery et al,[33] is increasingly prevalent in health and medical research and seems particularly appropriate for RETCs focused on environmental stress. Involving researchers, athletes, coaches, and other stakeholders in the collaborative planning and execution of RETCs ensures that the research remains relevant, practical, and addresses real-world challenges. To bridge the gap between research and applied practice, qualitative methods (eg, online surveys, interviews, and focus groups) can be used complementarily.[34] This integration of diverse perspectives supports evidence-driven decision making, enhances the RETC's capacity to manage environmental stress, and promotes better performance outcomes in a supportive training environment. Table 1 offers tips for maximizing the impact of an RETC.[32]

Conclusions

To move beyond controlled laboratory settings, RETCs offer valuable opportunities to improve practices through actionable research. However, they require meticulous planning with an emphasis on practicality, collaboration, and athlete-centered approaches. Successful delivery and evaluation depend on evidence-based practices and appropriate methodologies. Conducting interventions *with* athletes, rather than merely *on* them, can advance sport sciences, enhance performance, and refine evidence-based training strategies in high-performance environments through both knowledge translation and production. In the context of RETCs, the influence of the researcher–practitioner relationship on study outcomes must be considered. The Hawthorne effect suggests that a researcher's presence or feedback may alter coaches' decisions, or athletes' behaviors, complicating the distinction between the actual impact of the intervention and changes due to being observed.[35] A key challenge is gaining wider acceptance for this realistic research format within the academic community. Despite concerns about smaller sample sizes and less controlled conditions, we believe that the benefits for the sport-science community outweigh these drawbacks, a perspective not yet fully embraced by many academics and sport organizations.

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